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Path Integral formulation of Quantum Mechanics

Final Report

by

Tanuj Chauhan, PHYS1353

M.Sc. Physics IIT Guwahati

Project Supervisor: Prof. Justin R. David

IISc. Bangalore

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Contents

1	Path integral Generalization	1
1.1	Introduction	1
1.1.1	Classical Action - Principle of Least Action	1
1.2	Quantum Mechanical Amplitude	1
1.2.1	The Classical Limit	1
1.2.2	Path Integral - Sum Over All Paths	2
1.3	Quadratic lagrangian Kernel	3
1.4	Wavefunction	3
1.5	Wavefunction evolution with time	3
1.6	Dirac Delta Function time evolution	4
1.7	Propagator in Wave Mechanics	4
1.8	Path Integral through Schrödinger approach	5
2	Some exactly Solvable problems	6
2.1	Propogator for free Particle	6
2.1.1	Propogator using operator method	6
2.1.2	Free Particles using Feynman's approach	6
2.2	Propagator for Harmonic Oscillator	7
2.2.1	Using Operator method	7
2.2.2	Using Path integral approach	8
2.3	Forced Harmonic Oscillator	9
2.4	Propogator for Particle on a circle	10
2.5	Charged Particle moving in circular ring, with magnetic field passing through it	11
2.6	Particle in constant magnetic field	12
3	Relating the Kernel to physical quantity	14
3.1	Kernel and Partition Function	14
3.2	Kernel and $1/H$	14
3.3	$\text{Det}(H)$ and Kernel	14
3.4	Kernel and Ground State Energy	14
4	Many-body Systems	16
4.1	System of harmonic oscillators on 1D Circular ring	16
4.1.1	Quantization of this system	17
4.1.2	Partition Function	17
A	Action Problems	18
A.1	Free Particle Action	18
A.2	Action for Harmonic Oscillator	18
A.3	Particle under a constant force F	18

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Path integral formulation of Quantum Mechanics

Tanuj Chauhan under the guidance of Prof. Justin R. David¹

¹M.Sc. Physics at Department of Physics, Indian Institute of Technology, Guwahati

²Centre for High Energy Physics, Indian Institute of Science, Bangalore

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Abstract

The path integral formulation, introduced by Richard Feynman, provides a powerful alternative to the traditional Schrödinger and Heisenberg formulations in quantum mechanics. This project explores the path integral formulation and its connection with Schrödinger's wave mechanics. We analyze several simple systems using both the path integral and conventional operator approaches to compare their methodologies and results. Furthermore, we investigate how the path integral kernel correlates with fundamental physical observables extensively used in quantum mechanics.

Chapter 1

Path integral Generalization

1.1 Introduction

To start with path integrals we must start with action. We know that action is defined as

$$\int_{t_a}^{t_b} L(\dot{x}, x, t) dt \quad (1.1)$$

1.1.1 Classical Action - Principle of Least Action

Principle of least action in classical physics states that out of all possible paths between two points x_a at t_a and x_b at t_b , the $x_{cl}(t)$ is the classical path taken.

In classical physics only this path is of special importance but in quantum physics all paths need to be considered.

For a particle moving in 1D, position at any point is represented by x, t and $x(t)$ represents the path. If a particle starts from point x_a at t_a and ends up at x_b at t_b then action:

$$S = \int_{t_a}^{t_b} L(\dot{x}, x, t) dt \quad (1.2)$$

where L is the Lagrangian of the system which classically is given by:

$$L = \frac{m}{2} \dot{x}^2 - V(x, t) \quad (1.3)$$

To find the classical path we need to find the extremum of Action S by usual means. Assume the starting point and end points to be fixed, we get:

$$\delta S = \left[\delta x \frac{\partial L}{\partial x} \right]_{t_a}^{t_b} - \int_{t_a}^{t_b} \delta x \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} \right] dt \quad (1.4)$$

In this equation, the first part is zero as we assumed initial and final position to be same for all paths. Additionally, for the path to be an extremum $\delta S = 0$.

Therefore, what we are left with is the well-known Euler-Lagrange Equation which gives us the classical path.

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad (1.5)$$

1.2 Quantum Mechanical Amplitude

In quantum mechanics all the paths contribute equally in magnitude but differently in phase. Here, the probability amplitude is given by $K(b, a)$ for paths from point a to b and it is given by:

$$K(b, a) = \sum_{\text{all paths from } a \text{ to } b} \phi[x(t)] \quad (1.6)$$

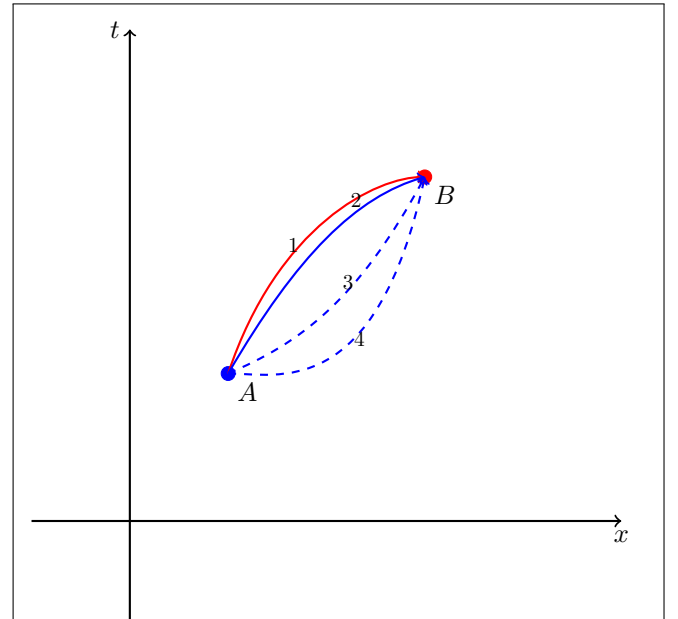


Figure 1.1: In this figure path 1 represents the classical path, and the path 2 just adjacent to it is a path for which S differs from S_{cl} by small amount and therefore it contributes to the classical path, but the paths 3 and 4, which have actions much different from the classical path, would not contribute.

1.2.1 The Classical Limit

Consider figure 1.1 if we change the path by a small value δx (i.e. small on classical scale) the change in S in scale of $\hbar$ will be huge. This change will lead to a

huge change in the phase and sine and cosine values will oscillate greatly between negative and positive values. The total contribution due to neighbouring paths with different action will be zero.

But if we consider neighbouring paths very close to x_{cl} such that S differs from S_{cl} by an amount less than $\hbar$, then it would contribute in the same phase and constructively interfere.

That's why we don't need to consider any other path which has different action S . The major contribution is only due to paths close to x_{cl} with

$$|S - S_{cl}| \sim \hbar \quad (1.7)$$

1.2.2 Path Integral - Sum Over All Paths

Since, now we know that in quantum mechanics all paths contribute, we sum the contribution of all paths between (x_a, t_a) and (x_b, t_b) .

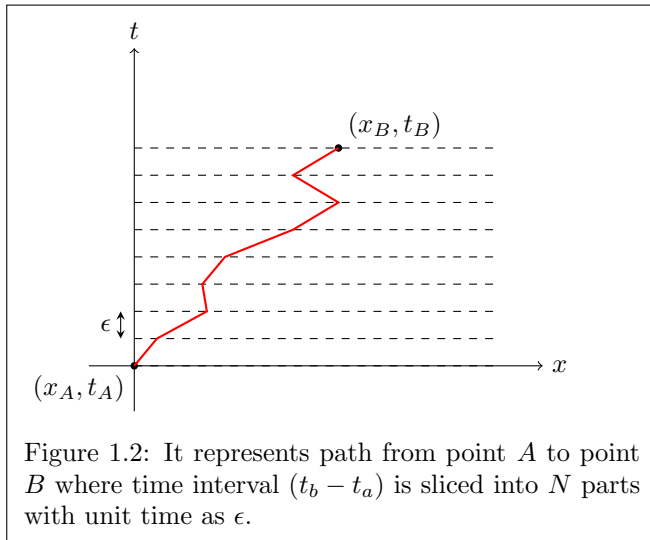
Time Slicing

First, let's divide the time interval $t_a - t_b$ into N small intervals, such that

$$\epsilon = \frac{t_a - t_b}{N} \quad (1.8)$$

$$t_0 = t_a, \quad t_N = t_b \quad (1.9)$$

$$x_0 = x_a, \quad x_N = x_b \quad (1.10)$$



So we may sum over all paths by taking multiple integrals over all x_i for i between 1 to $N-1$ as follows,

$$K(b, a) \sim \int \dots \int \int e^{\frac{i}{\hbar} S[x]} dx_1 dx_2 \dots dx_{N-1} \quad (1.11)$$

$$K(b, a) = A^{-N} \int \dots \int \int e^{\frac{i}{\hbar} S[x]} dx_1 dx_2 \dots dx_{N-1} \quad (1.12)$$

where,

$$S[b, a] = \int_{t_a}^{t_b} L(\dot{x}, x, t) dt \quad (1.13)$$

$$= \int_{t_1}^{t_2} L_1 dt + \int_{t_2}^{t_3} L_2 dt + \dots + \int_{t_{N-1}}^{t_N} L_{N-1} dt \quad (1.14)$$

This can be written as:

$$K(b, a) = \int_a^b e^{\frac{i}{\hbar} S[x]} \mathcal{D}x(t) \quad (1.15)$$

Events in Succession

The Rule of Two Events

$$S[b, a] = S[b, c] + S[c, a] \quad (1.16)$$

$$K(b, a) = \int_a^b e^{\frac{i}{\hbar} (S[b, c] + S[c, a])} \quad (1.17)$$

So we must be able to integrate over all paths from a to c , then over all the paths from c to b , and finally integrating over all possible values of x_c , therefore we get:

$$K(b, a) = \int_{-\infty}^{\infty} dx_c \int_c^b e^{\frac{i}{\hbar} S[b, c]} K(c, a) \mathcal{D}x(t) \quad (1.18)$$

$$K(b, a) = \int_{-\infty}^{\infty} dx_c K(b, c) K(c, a) \quad (1.19)$$

To Summarize: The kernel to go from $a \rightarrow b$ is the sum over all possible values of x_c , of the amplitude for particles to go from $a \rightarrow c$ then from $c \rightarrow b$.

The amplitude to go from $a \rightarrow c$ then to b is the kernel to go from a to c times the kernel to go from $c \rightarrow b$.

Amplitude for Events in Succession Similarly, if we have several events in succession between a and b , we can express the kernel as:

$$K(b, a) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx_c dx_d K(b, c) K(c, d) K(d, a) \quad (1.20)$$

Or if we have $N-1$ events between a and b then,

$$K(b, a) = \int_{-\infty}^{\infty} \int_{x_{N-1}} \dots \int_{x_2} \int_{x_1} K(b, N-1) \dots K(i+1, i) \quad (1.21)$$

$$\dots K(2, 1) K(1, a) dx_{N-1} \dots dx_2 dx_1 \quad (1.22)$$

Where,

$$K(i+1, i) = \exp^{\frac{i}{\hbar} \epsilon L\left(\frac{x_{i+1}-x_i}{2}, \frac{x_{i+1}+x_i}{2}, \frac{t_{i+1}+t_i}{2}\right)} dt \quad (1.23)$$

Or we can say the amplitude of a complete path to be the product of the amplitude of events in succession,

$$\phi[x(t)] = \lim_{\epsilon \rightarrow 0} \prod_{i=0}^{N-1} K(i+1, i) \quad (1.24)$$

1.3 Quadratic lagrangian Kernel

In general path integral equation 1.15 becomes difficult and lengthy to solve, but some special type of problems which have lagrangian of quadratic form can be solved in easier way using saddle point approximation. Here we will develop that method and apply it later in different problems. Suppose we have Lagrangian of form,

$$L = a(t\dot{x})^2 + b(t)\dot{x}x + c(t)x^2 + d(t)\dot{x} + e(t)x + f(t) \quad (1.25)$$

and we want to find,

$$K(x_b, t_b; x_a, t_a) = \int_a^b \mathcal{D}[x(t)] e^{\frac{i}{\hbar} S[x(t)]} \quad (1.26)$$

Suppose x_{cl} represent the classical path (extremum of action) then any path $x(t)$ can be represented in terms of classical path as;

$$x(t) = x_{cl} + y(t) \quad (1.27)$$

where $y(t)$ is the measure of deviation from classical path. Also,

$$\mathcal{D}[x(t)] = \mathcal{D}[y(t)] \quad (1.28)$$

as $x(t)$ at some time t is sum of classical path (constant) and deviation $y(t)$.

Since both classical path $x_{cl}(t)$ and $x(t)$ reach same endpoints deviation $y(t)$ at endpoints must be zero.

$$y(t_a) = y(t_b) = 0$$

Now using ?? we can write;

$$S[x(t)] = S[x_{cl}(t) + y(t)] \quad (1.29)$$

Now Since x_{cl} is extremum path $\frac{\partial S[x_{cl}]}{\partial x} = 0$, and therefore after applying saddle point approximation on $S[x(t)]$ we get;

$$S[x(t)] = S[x_{cl}] + \frac{1}{2}(x - x_{cl})^2 S''[x_{cl}] + \dots \quad (1.30)$$

and since we are considering quadratic lagrangian only we would be left with terms upto 2nd order;

$$S[x(t)] = S[x_{cl}] + \frac{1}{2}y^2 \frac{\partial^2 S[x_{cl}]}{\partial x^2} \quad (1.31)$$

Now substituting this $S[x(t)]$ in eq.1.26 we get,

$$K(x_b, t_b; x_a, t_a) = e^{\frac{i}{\hbar} S[x_{cl}]} \int_a^b \mathcal{D}[y(t)] \exp \left\{ \frac{i}{\hbar} \left(\frac{1}{2} \frac{\partial^2 S[x_{cl}]}{\partial x^2} \right) y^2 \right\} \quad (1.32)$$

$$K(x_b, t_b; x_a, t_a) = e^{\frac{i}{\hbar} S[x_{cl}]} F(T) \quad (1.33)$$

where,

$$F(t) = \exp \left\{ \frac{i}{\hbar} \left(\frac{1}{2} \frac{\partial^2 S[x_{cl}]}{\partial x^2} \right) y^2 \right\} \quad (1.34)$$

Therefore any 2nd order Lagrangian, we can find kernel directly using equation,

$$K(b, a) = e^{(i/\hbar) S_{cl}[b, a]} F(t_b, t_a) \quad (1.35)$$

1.4 Wavefunction

In path intergral approach to find the probability amplitude to reach a particle at some position in space and time we follow its path closely. But sometime it may be necessary to consider amplitude for particle at some position with unknown past.

Such an amplitude is given as Wavefunction $\psi(x, t)$.

Since its just another form of amplitude it must satisfy the rule of combination of 2 events in succession i.e. eq. 1.19

$$\psi(x_b, t_b) = \int_{-\infty}^{\infty} K(x_b, t_b; x_c, t_c) \psi(x_c, t_c) dx_b \quad (1.36)$$

It states that probability amplitude to reach at point x_b, t_b is the integral of product of amplitude of transition from $(x_c, t_c) \rightarrow (x_b, t_b)$ and probability amplitude to be at (x_c, t_c)

1.5 Wavefunction evolution with time

Now since we know about wave function we can solve a case where there is a given wavefunction and we will find how it evolve in time. Given a wavefunction

$$\psi(x_b, t = 0) = C e^{\frac{i}{\hbar} p x_b}$$

We have to find $\psi(x, t)$

We can find it using the relation 1.36,

$$\psi(X, T) = \int_{-\infty}^{\infty} K(X, T; x_b, 0) \psi(x_b, 0) dx_b \quad (1.37)$$

where K is free particle kernel,

$$K(X, T, x_b, 0) = \sqrt{\frac{m}{2\pi\hbar iT}} e^{\frac{im}{2\hbar} \frac{(X-x_b)^2}{T}} \quad (1.38)$$

Putting it in eq. 1.36 we get $\psi(X, T)$,

$$\psi(X, T) = \int_{-\infty}^{\infty} \left(C \sqrt{\frac{m}{2\pi\hbar i T}} e^{\frac{im}{2\hbar} \frac{(X-x_b)^2}{T} + \frac{i}{\hbar} p x_b} \right) dx_b \quad (1.39)$$

$$\Rightarrow C \sqrt{\frac{m}{2\pi\hbar i T}} \exp \left\{ \frac{im}{2\hbar T} \left(x_b^2 - 2x_b \left(X - \frac{Tp}{m} \right) + X^2 \right) \right\} \quad (1.40)$$

$$\Rightarrow C \sqrt{\frac{m}{2\pi\hbar i T}} \exp \left\{ \frac{im}{2\hbar T} \left(x_b - \left(X - \frac{Tp}{m} \right) \right)^2 \right\} \\ \times \exp \left\{ \frac{im}{2\hbar T} \left(X^2 - \left(X - \frac{Tp}{m} \right)^2 \right) \right\} \quad (1.41)$$

Which turns out to be of form of Gaussian integral and gives:

$$\psi(X, T) = C \sqrt{\frac{m}{2\pi\hbar i T}} \sqrt{\frac{2\pi\hbar i T}{m}} \exp \left\{ \frac{i}{\hbar} \left(pX - \frac{p^2}{2m} T \right) \right\} \quad (1.42)$$

$$\psi(X, T) = C \exp \left\{ \frac{i}{\hbar} \left(pX - \frac{p^2}{2m} T \right) \right\} \quad (1.43)$$

This result was expected as we know this addition term as unitary time evolution operator.

1.6 Dirac Delta Function time evolution

To get more idea about propagator from point of view of operator method we must take a look at how dirac delta function as wavefunction evolve with time in quantum mechanics.

Given $\psi(x) = \delta(x - x') = \langle x | x' \rangle$

and we know after time evolution new wavefunction will be given by,

$$\psi(x, t) = \langle x, t | x' \rangle = \langle x | e^{-\frac{i}{\hbar} H t} | x' \rangle \quad (1.44)$$

After inserting the completeness relation in this equation we get,

$$\psi(x, t) = \int_{-\infty}^{\infty} \langle x | e^{-\frac{i}{\hbar} H t} | p' \rangle \langle p' | x' \rangle dp' \quad (1.45)$$

Using free particle hamiltonian,

$$\psi(x, t) = \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} \frac{(p')^2}{2m} t} \langle x | p' \rangle \langle p' | x' \rangle dp' \quad (1.46)$$

$$\Rightarrow \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp' e^{-\frac{i}{\hbar} \frac{(p')^2}{2m} t} e^{ip(x-x')} \quad (1.47)$$

Solving this integral we get,

$$\Rightarrow \sqrt{\frac{m}{2\pi\hbar i t}} \exp \left(\frac{im}{2\hbar} \frac{(x - x')^2}{t} \right) \quad (1.48)$$

Here we notice that the evolved dirac delta function for free particle is same as propagator for free particle and it makes sense because while finding propagator we find the amplitude of particle to be at some position (x, t) in space and time by following its path from some past position (x_0, t_0) .

1.7 Propagator in Wave Mechanics

Now we will try to compare it to what we know through wave analysis of quantum mechanics.

In wave analysis of quantum mechanics if we are given an initial state $|\alpha, t_0\rangle$ it evolves in time as follows :

$$|\alpha, t_0; t\rangle = e^{-\frac{i\mathcal{H}t}{\hbar}} |\alpha, t_0\rangle \quad (1.49)$$

$$|\alpha, t_0; t\rangle = \sum_{a'} e^{-\frac{i\mathcal{H}t}{\hbar}} |a'\rangle \langle a' | \alpha, t_0 \rangle \quad (1.50)$$

Operating from position bra $\langle x' | \rightarrow$

$$\langle x' | \alpha, t_0; t\rangle = e^{-\frac{iE_{a'}t}{\hbar}} \langle x' | a' \rangle \langle a' | \alpha, t_0 \rangle \quad (1.51)$$

Since $\langle x' | \alpha, t_0; t\rangle = \psi_{\alpha}(t)$,

$$\psi_{\alpha}(t) = \psi(x', t) \quad (1.52)$$

$$= \sum_{a'} e^{-\frac{iE_{a'}t}{\hbar}} U_{a'} C_{a'} \quad (1.53)$$

where $C_{a'} = \langle a' | \alpha, t_0 \rangle$ and $U_{a'} = \langle x' | a' \rangle$

$$C_{a'} = \int d^3x U_{a'}^*(x) \psi(x', t_0) \quad (1.54)$$

and also by operating from position bra $\langle x'' | \rightarrow$

$$\langle x'' | \alpha, t_0 \rangle = \sum_{a'} \int e^{-\frac{iE_{a'}t}{\hbar}} \langle x'' | a' \rangle \langle a' | x' \rangle \langle x' | \alpha, t_0 \rangle \quad (1.55)$$

Comparing it to :

$$\psi(x'', t) = \int d^3x K(x'', t; x', t_0) \psi(x', t_0) \quad (1.56)$$

we get,

$$K(x'', t; x', t_0) = \sum_{a'} e^{-\frac{iE_{a'}(t-t_0)}{\hbar}} \langle x'' | a' \rangle \langle a' | x' \rangle \quad (1.57)$$

$$K(x'', t; x', t_0) = \langle x'' | e^{-\frac{i\mathcal{H}(t-t_0)}{\hbar}} | x' \rangle \quad (1.58)$$

$$K(x'', t; x', t_0) = \langle x'', t | x', t_0 \rangle \quad (1.59)$$

Therefore propagator can be written as transition amplitude.

1.8 Path Integral through Schrödinger approach

We know the transition amplitude is $K(x'', t; x', t_0) = \langle x'', t | x', t_0 \rangle$

By time slicing - dividing the time interval $(t - t_0)$ into N intervals such that;

$$\frac{(t - t_0)}{N} \quad (1.60)$$

$$\begin{aligned} \langle x'' | \left(e^{-\frac{iH}{\hbar} \frac{(t-t_0)}{N}} \right)^N | x' \rangle = \\ \int dx_{N-1} \dots \int dx_2 \int dx_1 \langle x'' | e^{-\frac{iH\epsilon}{\hbar}} | x_{N-1} \rangle \\ \times \langle x_{N-1} | e^{-\frac{iH\epsilon}{\hbar}} | x_{N-2} \rangle \dots \langle x_2 | e^{-\frac{iH\epsilon}{\hbar}} | x_1 \rangle \langle x_1 | e^{-\frac{iH\epsilon}{\hbar}} | x' \rangle \end{aligned}$$

Consider,

$$\langle x_{j+1} | e^{\frac{i\epsilon}{\hbar} (\frac{p^2}{2m} - V(x))} | x_j \rangle = e^{\frac{-i\epsilon}{\hbar} V(x_j)} \langle x_{j+1} | e^{\frac{-i\epsilon}{\hbar} (\frac{p^2}{2m})} | x_j \rangle \quad (1.61)$$

$$= e^{\frac{-i\epsilon}{\hbar} V(x_j)} \int dp'' dp' \langle x_{j+1} | p'' \rangle \langle p'' | e^{\frac{-i\epsilon}{\hbar} (\frac{p^2}{2m})} | p' \rangle \langle p' | x_j \rangle \quad (1.62)$$

Since,

$$\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{ipx}{\hbar}} \quad (1.63)$$

So the above equation will become,

$$= \frac{e^{\frac{-i\epsilon}{\hbar} V(x_j)}}{2\pi\hbar} \int dp' e^{\frac{-i}{\hbar} \left(\frac{p'^2}{2m} \epsilon - p'(x_{j+1} - x_j) \right)} \quad (1.64)$$

$$= \frac{e^{\frac{-i\epsilon}{\hbar} V(x_j)}}{2\pi\hbar} e^{\frac{im}{2\hbar} \frac{(x_{j+1} - x_j)^2}{\epsilon}} \int dp' e^{\frac{-i\epsilon}{2m\hbar} \left(p' - \frac{(x_{j+1} - x_j)}{\epsilon} \right)^2} \quad (1.65)$$

$$= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} e^{\frac{i}{\hbar} \sum_{j=0}^{N-1} \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_j) \right] \epsilon} \quad (1.66)$$

Placing it back into our main equation,

$$\int_{x'}^{x''} \mathcal{D}(x) e^{\frac{i}{\hbar} \int_a^b \mathcal{L}(\dot{x}, x, t) dt} = \int_a^b \mathcal{D}(x) e^{\frac{i}{\hbar} \mathcal{S}[a, b]} \quad (1.67)$$

This is the expression for the propagator in our approach of path integral.

Chapter 2

Some exactly Solvable problems

Now we will solve some standard problems in quantum mechanics that will help us relate two methods of finding the propagator and understand it better.

2.1 Propogator for free Particle

Lets start with the most simple case of finding propogator for free particle in 1D.

2.1.1 Propogator using operator method

1st we will start by using operator method as follows, We know for a free particle Hamiltonian $\mathcal{H} = \frac{P^2}{2m}$ So placing hamiltonian in propogator equation 1.59 we get:

$$K = \langle x'', t | x', t_0 \rangle = \langle x'' | e^{-\frac{iHt}{\hbar}} | x' \rangle \quad (2.1)$$

$$\Rightarrow \int \frac{dp'}{2\pi\hbar} \langle x'' | e^{-\frac{iP'^2(t-t_0)}{\hbar 2m}} | p' \rangle \langle p' | x' \rangle \quad (2.2)$$

$$\Rightarrow \int \frac{dp'}{2\pi\hbar} e^{-\frac{ip'^2(t-t_0)}{\hbar 2m}} e^{\frac{i}{\hbar} p' \cdot x'} \langle x'' | p' \rangle \langle p' | x' \rangle \quad (2.3)$$

$$\Rightarrow \int \frac{dp'}{2\pi\hbar} e^{-\frac{ip'^2}{\hbar 2m}(t-t_0) - \frac{i}{\hbar} p' \cdot (x'' - x')} \quad (2.4)$$

$$\Rightarrow \frac{1}{2\pi\hbar} e^{\frac{im}{2\hbar} \frac{(x'' - x')^2}{(t-t_0)}} \int_{-\infty}^{\infty} dp' e^{-\frac{ip'^2}{2m\hbar} \left[p' - m \left(\frac{x'' - x'}{t-t_0} \right) \right]} \quad (2.5)$$

$$K = \sqrt{\frac{m}{2\pi\hbar i(t-t_0)}} e^{\frac{im}{2\hbar} \frac{(x'' - x')^2}{(t-t_0)}} \quad (2.6)$$

Eq. 2.6 gives us the propogator for free particle in 1D.

Now lets try to get the same result using feynman's approach.

2.1.2 Free Particles using Feynman's approach

To get propogator through path integral approach we need to 1st find classical action. From equation A.4 we know that action for free particle is:

$$\Rightarrow S = \frac{m}{2} \frac{(x_b - x_a)^2}{(t_b - t_a)}$$

Using equation 1.12 we get kernel K for free particle as:

$$K_0(b, a) = \lim_{\epsilon \rightarrow 0} \left(\frac{m}{2\pi i \hbar} \right)^{N/2} \int \dots \int \exp \left\{ \frac{im}{2\hbar\epsilon} \sum_{i=1}^N (x_i - x_{i-1})^2 \right\} dx_1 \dots dx_N \quad (2.7)$$

Where ϵ is the unit of time we get after slicing time interval into N parts i.e.

$$\epsilon = \frac{(t_b - t_a)}{N} = \frac{T}{N} \quad (2.8)$$

Since all integrals here are of the form:

$$\Rightarrow \int_{-\infty}^{\infty} \exp \left[i \left\{ Q(x'' - x_c)^2 + V(x_c - x')^2 \right\} \right] = \mathcal{I} \quad (2.9)$$

This integral simplifies to:

$$\mathcal{I} = \sqrt{\frac{i\pi}{Q+V}} \exp \left\{ i \left(-\frac{(Qx'' + Vx')^2}{Q+V} + Qx''^2 + Vx'^2 \right) \right\} \quad (2.10)$$

Using above integral we can find:

$$\Rightarrow A^{-2} \int_{-\infty}^{\infty} \exp \left\{ \frac{im}{2\hbar\epsilon} ((x_2 - x_1)^2 + (x_1 - x_0)^2) \right\} dx_1 \quad (2.11)$$

Comparing it to equation 2.10 we have

$$Q = V = \frac{m}{2\hbar\epsilon}$$

where, $x'' = x_2$, $x' = x_0$ & $x_c = x_1$

Solving this we would get

$$\Rightarrow A^{-2} \sqrt{\frac{i\pi\hbar\epsilon}{m}} \exp \left\{ \frac{im}{4\hbar\epsilon} (x_2 - x_0)^2 \right\} \quad (2.12)$$

After solving these integrals N-1 times we would get our result as:

$$\Rightarrow A^{-N} \left(\frac{2\pi\hbar\epsilon i}{m} \right)^{\frac{N}{2}} \sqrt{\frac{m}{2\pi i\hbar(N\epsilon)}} \exp \left\{ \frac{im(x_N - x_0)^2}{2\hbar(N\epsilon)} \right\} \quad (2.13)$$

Since from equation 2.8 $N\epsilon = T$ we get:

$$K(b, a) = A^{-N} \left(\frac{2\pi\hbar\epsilon i}{m} \right)^{\frac{N}{2}} \sqrt{\frac{m}{2\pi i\hbar T}} \times \exp \left\{ \frac{im(x_N - x_0)^2}{2\hbar T} \right\} \quad (2.14)$$

Which is similar to the answer we got in previous subsection.

Using Quadratic Lagrangian Property

Since the Lagrangian for free particle is quadratic in nature we may also use the approach we discussed in section 1.3.

Equation 1.35 in section 1.3 says,

$$K(b, a) = e^{(i/\hbar)S_{cl}[b,a]} F(t_b, t_a) \quad (2.15)$$

Since we know the classical action for free particle from equation A.4, we can write above equation as:

$$K(b, a) = F(t_b, t_a) \exp \left\{ \frac{im(x_b - x_a)^2}{2\hbar(t_b - t_a)} \right\} \quad (2.16)$$

Now we must figure out $F(t_b, t_a)$, for that lets say we consider a path which is deviated from classical path by $y(t)$, i.e.,

$$x(t) = x_{cl}(t) + y(t) \quad (2.17)$$

So action for this path can be expressed as:

$$S[x(t)] = S_{cl}[b, a] + \int_{t_a}^{t_b} \frac{m}{2} \dot{y}^2 dt \quad (2.18)$$

From this we can find $K(b, a)$ as,

$$K(b, a) = \exp \left\{ \frac{im(x_b - x_a)^2}{2\hbar(t_b - t_a)} \right\} F(t_b, t_a) \quad (2.19)$$

where $F(t_b, t_a)$ turns out to be:

$$F(t_b, t_a) = \int_0^1 \exp \left\{ \frac{i}{\hbar} \int_{t_a}^{t_b} \frac{m}{2} \dot{y}^2 dt \right\} \mathcal{D}y(t) \quad (2.20)$$

The limit in the integral goes from 0 to 0 because the boundary points i.e. $x(t_a) = x_{cl}(t_a)$ and $x(t_b) = x_{cl}(t_b)$ remains same. or $y(t_a) = y(t_b) = 0$

To solve the integral for $F(t_b, t_a)$ we will use the rule of two events i.e. equation 1.19

So,

$$K(0, T; 0, 0) = \int_{-\infty}^{\infty} dx K(0, T; x, t) K(x, t; 0, 0) \quad (2.21)$$

Solving this we get,

$$F(T, 0) = \int_{-\infty}^{\infty} F(T - t) \exp \left(\frac{imx^2}{2\hbar(T - t)} \right) F(t) \exp \left(\frac{imx^2}{2\hbar t} \right) dx \quad (2.22)$$

$$\Rightarrow F(T) = F(T - t) F(t) \int_{-\infty}^{\infty} \exp \left(\frac{imx^2 T}{2\hbar(T - t)t} \right) \quad (2.23)$$

Solving it we get,

$$\frac{F(T)}{F(T - t)F(t)} = \sqrt{\frac{2\pi\hbar i(T - t)t}{mT}} \quad (2.24)$$

From which we can say that $F(T)$ is,

$$F(T) = \sqrt{\frac{m}{2\pi\hbar iT}} \quad (2.25)$$

Substituting it in 2.16 we get,

$$K = \sqrt{\frac{m}{2\pi\hbar i(t - t_0)}} e^{\frac{im}{2\hbar} \frac{(x'' - x')^2}{(t - t_0)}} \quad (2.26)$$

2.2 Propagator for Harmonic Oscillator

Now for the case of Simple Harmonic Oscillator we will try to get propagator by both ways.

2.2.1 Using Operator method

For a Simple Harmonic Oscillator Hamiltonian is:

$$\mathcal{H} = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2} \quad (2.27)$$

Considering Creation and annihilation operator,

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{ip}{m\omega} \right) \quad (2.28)$$

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{ip}{m\omega} \right) \quad (2.29)$$

$$a^\dagger a = N = \frac{H}{\hbar\omega} - \frac{1}{2} \quad (2.30)$$

$$H = \left(N + \frac{1}{2} \right) \hbar\omega$$

Wave function for simple harmonic Oscillator is,

$$\psi = \langle x|n \rangle = \frac{1}{\pi^{1/4} \sqrt{2^n n!}} \frac{1}{\sqrt{x_0}} H_n \left(\frac{x}{x_0} \right) e^{-\frac{x^2}{2x_0^2}} \quad (2.31)$$

Where H_n is Hermite polynomial and $x_0 = \sqrt{\frac{\hbar}{m\omega}}$ is the position ground state.

Now propagator is given as,

$$K = \langle x_b, t | x_a, t_0 \rangle \quad (2.32)$$

$$\begin{aligned} &= \langle x_b | e^{-\frac{i}{\hbar} H(t-t_0)} | x_a \rangle \\ &= \sum_n \langle x_b | e^{-\frac{i}{\hbar} H(t-t_0)} | n \rangle \langle n | x_a \rangle \\ &= \sum_n \langle x_b | n \rangle \langle n | x_a \rangle e^{-\frac{i}{\hbar} (n + \frac{1}{2}) \hbar \omega (t-t_0)} \\ &= \sum_n \psi_n(x_b) \psi_n^*(x_a) e^{-\frac{i}{\hbar} (n + \frac{1}{2}) \hbar \omega (t-t_0)} \end{aligned} \quad (2.33)$$

$$\begin{aligned} &= \sum_n \frac{1}{\sqrt{\pi} 2^n n!} \frac{1}{x_0} e^{-\frac{x_b^2 + x_a^2}{2x_0^2}} H_n \left(\frac{x_b}{x_0} \right) \\ &\quad \times H_n \left(\frac{x_a}{x_0} \right) e^{-\frac{i}{\hbar} (n + \frac{1}{2}) \hbar \omega (t-t_0)} \end{aligned} \quad (2.34)$$

Here we will be using the identity of sum over product of 2 hermite polynomial,

$$\begin{aligned} \sum_n \frac{c^n}{2^n n!} H_n(a) H_n(b) &= \frac{\exp(a^2 + b^2)}{\sqrt{1 - c^2}} \\ &\quad \times \exp \left(-\frac{(a^2 + b^2 - 2ab)}{(1 - c^2)} \right) \end{aligned} \quad (2.35)$$

Let $c = e^{-i\omega(t-t_0)}$ then,

$$\begin{aligned} \frac{c}{1 - c^2} &= \frac{e^{-i\omega(t-t_0)}}{1 - e^{-2i\omega(t-t_0)}} \\ &= \frac{1}{e^{i\omega(t-t_0)} - e^{-i\omega(t-t_0)}} \\ &= \frac{1}{2i \sin \omega(t-t_0)} \end{aligned} \quad (2.36)$$

and,

$$\frac{1 + c^2}{1 - c^2} = \frac{\cos \omega(t-t_0)}{i \sin \omega(t-t_0)} \quad (2.37)$$

Therefore using 2.35, we can write 2.34 as:

$$\begin{aligned} K &= \frac{1}{\sqrt{\pi} x_0} e^{-\frac{x_b^2 + x_a^2}{2x_0^2}} \frac{c^{1/2}}{\sqrt{1 - c^2}} e^{\frac{x_b^2 + x_a^2}{x_0^2}} \\ &\quad \times \exp \left[-\frac{x_a^2 + x_b^2 - 2x_a x_b c}{x_0^2 (1 - c^2)} \right] \end{aligned} \quad (2.38)$$

$$\begin{aligned} &= \frac{1}{\sqrt{\pi} x_0} \frac{c^{1/2}}{\sqrt{1 - c^2}} \\ &\quad \times \exp \left[-\frac{(x_a^2 + x_b^2)(1 + c^2) + 2(2x_a x_b c)}{2x_0^2 (1 - c^2)} \right] \end{aligned} \quad (2.39)$$

Now after substituting $\frac{c}{1 - c^2}$ and $\frac{1 + c^2}{1 - c^2}$ we get the kernel as,

$$\begin{aligned} K &= \frac{1}{\sqrt{\pi}} \sqrt{\frac{m\omega}{\hbar}} \sqrt{\frac{\csc \omega(t-t_0)}{2i}} \\ &\quad \times \exp \left[-\frac{i \csc \omega(t-t_0)}{2x_0^2} \{2x_b x_a - (x_b^2 + x_a^2) \cos \omega(t-t_0)\} \right] \\ K &= \sqrt{\frac{m\omega \csc \omega(t-t_0)}{2\pi i \hbar}} \\ &\quad \times \exp \left[-\frac{im\omega \csc \omega(t-t_0)}{2\hbar} \{2x_b x_a - (x_b^2 + x_a^2) \cos \omega(t-t_0)\} \right] \end{aligned} \quad (2.40)$$

2.2.2 Using Path integral approach

Since we know that for Harmonic oscillator classical action is given by equation A.17 i.e.

$$S_{cl} = \frac{m\omega}{2 \sin(\omega T)} [(x_b^2 + x_a^2) \cos(\omega T) - 2x_a x_b]$$

Now using equation 1.35 we may write kernel for harmonic oscillator as:

$$K(b, a) = e^{(i/\hbar) \frac{m\omega}{2 \sin(\omega T)} [(x_b^2 + x_a^2) \cos(\omega T) - 2x_a x_b]} [b, a] F(T) \quad (2.41)$$

Similar to the method we used earlier for free particle we can find $F(T)$ as follows,

$$K(0, t + s; 0, 0) = \int_{-\infty}^{\infty} dx K(0, t + s; x, s) K(x, s; 0, 0) \quad (2.42)$$

$$F(T) = \int_{-\infty}^{\infty} dx F(t) F(s) \exp \left\{ \frac{im\omega x^2}{2\hbar} \left(\frac{\cos \omega s}{\sin \omega s} + \frac{\cos \omega t}{\sin \omega t} \right) \right\} \quad (2.43)$$

Solving this we get,

$$\frac{F(t+s)}{F(t)F(s)} = \int_{-\infty}^{\infty} dx \exp \left(\frac{im\omega \sin \omega(t+s)}{2\hbar \sin \omega s \sin \omega t} x^2 \right) \quad (2.44)$$

$$\frac{F(t+s)}{F(t)F(s)} = \sqrt{\frac{2\pi\hbar \sin \omega t \sin \omega s}{m\omega \sin \omega(t+s)}} \quad (2.45)$$

Which gives $F(T)$ as,

$$\Rightarrow F(T) = \sqrt{\frac{m\omega}{2\pi\hbar i \sin \omega T}} \quad (2.46)$$

and finally Kernel is,

$$K(b, a) = \sqrt{\frac{m\omega}{2\pi\hbar i \sin \omega T}} e^{(i/\hbar) \frac{m\omega}{2 \sin(\omega T)} [(x_b^2 + x_a^2) \cos(\omega T) - 2x_a x_b]} [b, a] \quad (2.47)$$

Therefore we can say that Path integral approach can be used to solve the problem of finding propagator and it agrees with the solution obtained by operator method approach.

Now we will solve some more problems to get better understanding of this concept.

2.3 Forced Harmonic Oscillator

We know the lagrangian for forced harmonic oscillator is given as:

$$\mathcal{L} = \frac{1}{2}(m\dot{x}^2 - m\omega^2 x^2) + f(t)x \quad (2.48)$$

$$S_{cl} = \int_{-\infty}^{\infty} \mathcal{L} dt \quad (2.49)$$

Using 1.5 we get equation of motion as:

$$\ddot{x} = -\omega^2 + g(t) \quad (2.50)$$

where $g(t) = f(t)/m$,

Now,

$$\int_{t_a}^{t_b} \dot{x}^2 dt = \dot{x}|_{t_a}^{t_b} - \int_{t_a}^{t_b} \ddot{x} x dt \quad (2.51)$$

Substituting $\ddot{x}$ in above equation,

$$\int_{t_a}^{t_b} \dot{x}^2 dt = \dot{x}|_{t_a}^{t_b} - \int_{t_a}^{t_b} \left(-\omega^2 x + \frac{f(t)}{m} \right) x dt \quad (2.52)$$

On solving this and collecting the terms of action on LHS we get,

$$S_{cl} = \int_{t_a}^{t_b} \frac{m}{2} \dot{x}^2 dt + \int_{t_a}^{t_b} f(t)x dt \quad (2.53)$$

For the equation of motion (i.e. 2.50) we can say that for the homogeneous part solution must be of the form,

$$x_c = A \sin \omega(t - t_a) + B \sin \omega(t_b - t) \quad (2.54)$$

Now to find general solution for the non homogeneous equation 2.50 we will use the following method of (variation of parameter),

1st we assume that,

$$x = x_c + x_p \quad (2.55)$$

$$x_p(t) = x_1 u_1 + x_2 u_2 \quad (2.56)$$

where, $x_1 = \sin \omega(t - t_a)$ & $x_2 = \sin \omega(t_b - t)$
 u_1 and u_2 can be calculated as follows:

$$u_1 = - \int \frac{x_2 g(t)}{W(x_2, x_1)} \quad (2.57)$$

and,

$$u_2 = \int \frac{x_1 g(t)}{W(x_2, x_1)} \quad (2.58)$$

Where $W(x_1, x_2)$ is the 'Wronskian' which can be calculated as:

$$W(x_1, x_2) = x_1 x_2' - x_2 x_1' \quad (2.59)$$

For our case:

$$W(x_1, x_2) = -\omega \sin(t_b - t_a) \quad (2.60)$$

therefore x_p will be,

$$x_p(t) = - \frac{\sin \omega(t - t_a)}{m\omega \sin \omega(t_b - t_a)} \int_t^{t_b} \sin \omega(t_b - t') f(t') dt' \\ - \frac{\sin \omega(t_b - t)}{m\omega \sin \omega(t_b - t_a)} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \quad (2.61)$$

Since at end points x_p must be zero and satisfy boundary conditions,

$$x_p(t_a) = x_p(t_b) = 0$$

So at $t = t_a$, $x = x_a$ i.e.

$$x_a = B \sin \omega(t_b - t_a)$$

or,

$$B = \frac{x_a}{\sin \omega(t_b - t_a)} \quad (2.62)$$

Similarly,

$$x_b = A \sin \omega(t_b - t_a)$$

or,

$$A = \frac{x_b}{\sin \omega(t_b - t_a)} \quad (2.63)$$

now using eq. 2.55 and 2.61 we get the general solution as:

$$x = x_b \frac{\sin \omega(t - t_a)}{\sin \omega T} + x_a \frac{\sin \omega(t_b - t)}{\sin \omega T} \\ - \frac{\sin \omega(t - t_a)}{m\omega \sin \omega(t_b - t_a)} \int_t^{t_b} \sin \omega(t_b - t') f(t') dt' \\ - \frac{\sin \omega(t_b - t)}{m\omega \sin \omega(t_b - t_a)} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \quad (2.64)$$

and,

$$\dot{x} = \frac{\omega}{\sin \omega T} (x_b \cos \omega(t - t_a) - x_a \cos \omega(t_b - t)) \\ - \frac{\omega \cos \omega(t - t_a)}{m\omega \sin \omega T} \int_t^{t_b} \sin \omega(t_b - t') f(t') dt' \\ + \frac{\omega \cos \omega(t_b - t)}{m\omega \sin \omega T} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \quad (2.65)$$

Now,

$$\dot{x}|_{t_a}^{t_b} = \frac{\omega}{\sin \omega T} [(x_b^2 + x_a^2) \cos \omega T - 2x_a x_b] \\ + \frac{x_b}{m\omega} \int_{t_a}^{t_b} \sin \omega(t' - t_a) f(t') \\ + \frac{x_a}{m\omega} \int_{t_a}^{t_b} \sin \omega(t_b - t') f(t') \quad (2.66)$$

and 2nd term of S_{cl} is,

$$\begin{aligned} \int_{t_a}^{t_b} x f(t) &= x_b \int_{t_a}^{t_b} f(t) \frac{\sin \omega(t - t_a)}{\sin \omega T} dt \\ &\quad + x_a \int_{t_a}^{t_b} f(t) \frac{\sin \omega(t_b - t)}{\sin \omega T} dt \\ &- \int_{t_a}^{t_b} \frac{f(t) dt}{m\omega} \frac{\sin \omega(t - t_a)}{\sin \omega T} \int_t^{t_b} \sin \omega(t_b - t') f(t') dt' \\ &- \int_{t_a}^{t_b} \frac{f(t) dt}{m\omega} \frac{\sin \omega(t_b - t)}{\sin \omega T} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \end{aligned} \quad (2.67)$$

Substituting both in eq. 2.53 we get:

$$\begin{aligned} S_{cl} &= \frac{m\omega}{2 \sin \omega T} [(x_b^2 + x_a^2) \cos \omega T - 2x_a x_b \\ &\quad + \frac{2x_b}{m\omega} \int_{t_a}^{t_b} \sin \omega(t' - t_a) f(t') \\ &\quad + \frac{2x_a}{m\omega} \int_{t_a}^{t_b} \sin \omega(t_b - t') f(t')] \\ &- \int_{t_a}^{t_b} \frac{f(t) dt}{2m\omega} \frac{\sin \omega(t - t_a)}{\sin \omega T} \int_t^{t_b} \sin \omega(t_b - t') f(t') dt' \\ &- \int_{t_a}^{t_b} \frac{f(t) dt}{2m\omega} \frac{\sin \omega(t_b - t)}{\sin \omega T} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \end{aligned} \quad (2.68)$$

$$\quad (2.69)$$

Now the integral:

$$\int_{t_a}^{t_b} \int_t^{t_b} \frac{f(t)}{2m\omega} \frac{\sin \omega(t - t_a)}{\sin \omega T} \sin \omega(t_b - t') f(t') dt' dt$$

by changing limits can be expressed as:

$$\Rightarrow \int_{t_a}^{t_b} \int_{t_a}^{t'} \frac{f(t)}{2m\omega} \frac{\sin \omega(t - t_a)}{\sin \omega T} \sin \omega(t_b - t') f(t') dt dt'$$

Since t and t' are dummy variables they can be interchanged, Therefore we get:

$$\begin{aligned} \int_{t_a}^{t_b} \int_t^{t_b} \frac{f(t)}{2m\omega} \frac{\sin \omega(t - t_a)}{\sin \omega T} \sin \omega(t_b - t') f(t') dt' dt \\ = \int_{t_a}^{t_b} \int_{t_a}^t \frac{f(t')}{2m\omega} \frac{\sin \omega(t' - t_a)}{\sin \omega T} \sin \omega(t_b - t) f(t) dt' dt \end{aligned} \quad (2.70)$$

Substituting it back in S_{cl} we get,

$$\begin{aligned} S_{cl} &= \frac{m\omega}{2 \sin \omega T} [(x_b^2 + x_a^2) \cos \omega T - 2x_a x_b \\ &\quad + \frac{2x_b}{m\omega} \int_{t_a}^{t_b} \sin \omega(t' - t_a) f(t') \\ &\quad + \frac{2x_a}{m\omega} \int_{t_a}^{t_b} \sin \omega(t_b - t') f(t')] \end{aligned} \quad (2.71)$$

$$- 2 \int_{t_a}^{t_b} \frac{f(t) dt}{2m\omega} \frac{\sin \omega(t_b - t)}{\sin \omega T} \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt' \quad (2.72)$$

or,

$$\begin{aligned} S_{cl} &= \frac{m\omega}{2 \sin \omega T} [(x_b^2 + x_a^2) \cos \omega T - 2x_a x_b \\ &\quad + \frac{2x_b}{m\omega} \int_{t_a}^{t_b} \sin \omega(t - t_a) f(t) \\ &\quad + \frac{2x_a}{m\omega} \int_{t_a}^{t_b} \sin \omega(t_b - t) f(t) \\ &\quad - \frac{2}{m^2 \omega^2} \int_{t_a}^{t_b} f(t) dt \sin \omega(t_b - t) \int_{t_a}^t \sin \omega(t' - t_a) f(t') dt'] \end{aligned} \quad (2.73)$$

Now using eq. 1.35 we can write kernel as:

$$K(b, a) = F(b, a) e^{(i/\hbar) S_{cl}} \quad (2.74)$$

Here $F(b, a)$ must be same as that for Simple harmonic oscillator as we saw after using saddle point approximation in eq.1.34 $F(T)$ depends only on quadratic term in x and additional term in forced oscillator lagrangian is linear, so it wouldn't contribute to $F(T)$.

$$F(b, a) = \sqrt{\frac{m\omega}{2\pi\hbar i \sin \omega T}} \quad (2.75)$$

Therefore,

$$K(b, a) = \sqrt{\frac{m\omega}{2\pi\hbar i \sin \omega T}} e^{(i/\hbar) S_{cl}} \quad (2.76)$$

2.4 Propogator for Particle on a circle

For a particle on a circle wavefunction is periodic and turns out to be of the form,

$$\psi = \frac{1}{\sqrt{2\pi}} e^{ik\theta} = \langle \theta | k \rangle \quad (2.77)$$

where k is an integer.

We know that propogator is given as:

$$K(\theta_b, t; \theta_a, t_0) = \langle \theta_b, t | \theta_a, t_0 \rangle \quad (2.78)$$

$$\Rightarrow \langle \theta_b | e^{\frac{i}{\hbar} H(t-t_0)} | \theta_a \rangle \quad (2.79)$$

$$\Rightarrow \sum_{k'=-\infty}^{\infty} \langle \theta_b | e^{\frac{i}{\hbar} H(t-t_0)} | k' \rangle \langle k' | \theta_a \rangle$$

$$\Rightarrow \sum_{k'=-\infty}^{\infty} \langle \theta_b | e^{\frac{i}{\hbar} \frac{\hbar^2 k'^2}{2m} (t-t_0)} | k' \rangle \langle k' | \theta_a \rangle$$

$$K(b, a) = \frac{1}{2\pi} \sum_{k' \in \mathbb{Z}} e^{\frac{i}{\hbar} \frac{\hbar^2 k'^2}{2m} (t-t_0) + i k' (\theta_b - \theta_a)} \quad (2.80)$$

This equation can be expressed using Poisson's Summation formula as,

$$K(b, a) = \sqrt{\frac{m}{2\pi i \hbar (t - t_0)}} \sum_{K=-\infty}^{\infty} \exp \left(\frac{im}{2\hbar} \frac{(\theta_b - \theta_a - 2\pi K)^2}{t - t_0} \right) \quad (2.81)$$

If we look closely at eq. 2.80 and eq. 2.81 we will notice that $K(b,a)$ is of the form of **Jacobi Theta function**.

We know jacobi theta function is defined as,

$$\vartheta(x, z) = \sum_{n \in \mathbb{Z}} e^{\pi n^2 x + 2\pi i n z} \quad (2.82)$$

and if we consider,

$$\theta(x) = \vartheta(ix, z=0) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x} \quad (2.83)$$

it turns out that after poisson's summation it gives us the relation:

$$\theta(x) = \frac{1}{\sqrt{x}} \theta\left(\frac{1}{x}\right) \quad (2.84)$$

where,

$$\theta\left(\frac{1}{x}\right) = \sum_{k \in \mathbb{Z}} e^{-\frac{\pi k^2}{x}} \quad (2.85)$$

If we compare with our result it turns out that the result we obtained in eq. 2.81 and 2.80 is jacobi theta function.

2.5 Charged Particle moving in circular ring, with magnetic field passing through it

We know for a circular path wave function is given as:

$$\psi_0 = \frac{1}{\sqrt{2\pi R}} e^{in\theta} = \langle \theta | n \rangle$$

But due to presence of magnetic potential in the neighbourhood of path of the particle we must apply Gauge Transformation,

Since B does not pass the path,

$$\begin{aligned} A &\rightarrow A' + \nabla \Lambda \\ A' &= \nabla \Lambda \end{aligned} \quad (2.86)$$

and,

$$\psi_0 \rightarrow \psi' = e^{\frac{iq\Lambda}{\hbar c}} \psi_0 \quad (2.87)$$

To find the value of A and Λ we will use the cylindrical coordinates with (r, θ, z)

$$\oint A' \cdot dl = \int B \cdot ds = \Phi \quad (2.88)$$

$$A' = \frac{\Phi}{2\pi R} \hat{\theta} \quad (2.89)$$

and since $A' = \nabla \Lambda$,

$$\begin{aligned} \nabla \Lambda &= \frac{\Phi}{2\pi R} \hat{\theta} \\ \frac{1}{R} \frac{\partial}{\partial \theta} \Lambda &= \frac{\Phi}{2\pi} \\ \Lambda &= \frac{\Phi}{2\pi} \theta \end{aligned} \quad (2.90)$$

Now Hamiltonian for particle in A' is :

$$H = \frac{1}{2m} \left(-i\hbar \nabla - \frac{qA'}{c} \right)^2 \quad (2.91)$$

Now using Schrödinger eq. $H\psi' = E\psi'$

$$\begin{aligned} \frac{1}{2m} \left(-\frac{i\hbar}{R} \frac{\partial}{\partial \theta} - \frac{qA'}{c} \right)^2 \psi' &= E\psi' \\ \frac{1}{2mR^2} \left(-i\hbar \frac{\partial}{\partial \theta} - \frac{q\Phi}{2\pi c} \right)^2 \psi' &= E\psi' \\ \frac{1}{2mR^2} \left(-\hbar^2 \frac{\partial^2}{\partial \theta^2} \psi' + \left(\frac{q\Phi}{2\pi c} \right)^2 \psi' + 2 \left(\frac{q\Phi}{2\pi c} \right) i\hbar \frac{\partial}{\partial \theta} \psi' \right) &= E\psi' \end{aligned} \quad (2.92)$$

Now we know that,

$$\begin{aligned} \psi' &= \exp \left(\frac{iq\Lambda}{\hbar c} + in\theta \right) \\ &\Rightarrow \exp \left(\frac{iq}{\hbar c} \frac{\Phi}{2\pi} + in\theta \right) \\ &\Rightarrow \exp \left(i \left(\frac{\Phi}{\Phi_0} + n \right) \theta \right) \end{aligned} \quad (2.93)$$

where $\Phi_0 = \frac{2\pi\hbar c}{q}$ and $l = \frac{\Phi}{\Phi_0}$

$$\frac{\partial \psi'}{\partial \theta} = in\psi' \quad (2.94)$$

$$\frac{\partial^2 \psi'}{\partial \theta^2} = -n^2 \psi' \quad (2.95)$$

placing these in Hamiltonian we get,

$$E_n = \frac{\hbar^2}{2mR^2} \left(n - \frac{q\Phi}{2\pi\hbar c} \right)^2 \quad (2.96)$$

$$E_n = \frac{\hbar^2}{2mR^2} \left(n - \frac{\Phi}{\Phi_0} \right)^2 \quad (2.97)$$

$$E_n = \frac{\hbar^2}{2mR^2} (n - l)^2 \quad (2.98)$$

Now Kernel is given as:

$$\begin{aligned} K(b, a) &= \langle \theta_b, t | \theta_a, t_0 \rangle \\ &\Rightarrow \langle \theta_b | e^{-\frac{i}{\hbar} H(t-t_0)} | \theta_a \rangle \\ &\Rightarrow \sum_{n'} \langle \theta_b | n' \rangle \langle n' | \theta_a \rangle \\ &= \frac{1}{2\pi R} \sum_{n'} e^{-\frac{i}{\hbar} E_n(t-t_0)} e^{i \left(\frac{q}{\hbar c} \frac{\Phi}{2\pi R} + n' \right) (\theta_b - \theta_a)} \\ &\Rightarrow \frac{1}{2\pi R} \sum_{n'} e^{-\frac{i}{\hbar} \left(\frac{\hbar^2}{2mR^2} \right) \left(n' - \frac{\Phi}{\Phi_0} \right)^2 (t-t_0) + i \left(\frac{\Phi}{\Phi_0} + n' \right) (\theta_b - \theta_a)} \\ &\Rightarrow \frac{1}{2\pi R} \sum_{n'} e^{-\frac{i}{\hbar} \left(\frac{\hbar^2}{2mR^2} \right) (n' - l)^2 (t-t_0) + i(l + n') (\theta_b - \theta_a)} \end{aligned} \quad (2.100)$$

It can be further simplified using Poisson's summation formula,

$$K = \sqrt{\frac{m}{2\pi\hbar(t-t_0)}} \sum_{K \in \mathbb{Z}} e^{i\left(\frac{mR^2}{2\hbar(t-t_0)}(\theta_b - \theta_a - 2\pi K)^2 + 2\pi K \frac{\Phi}{\Phi_0}\right)} \quad (2.101)$$

This result also is of the form of **Jacobi Theta function** eq. 2.85.

2.6 Particle in constant magnetic field

Suppose we are given a particle in constant magnetic field with Lagrangian:

$$\mathcal{L} = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + \frac{eB}{2c}(xy - \dot{x}y) \quad (2.102)$$

Since it is quadratic in nature we can use path integral approach to find propagator as follows;

Let $\omega = \frac{eB}{mc}$ From eq. 1.5 we can get the equations of motion as:

$$\begin{aligned} \ddot{x} &= \omega \dot{y} \\ \ddot{y} &= -\omega \dot{x} \\ \ddot{z} &= 0 \end{aligned} \quad (2.103)$$

Now to calculate action:

$$S_{cl} = \int \mathcal{L} dt \quad (2.104)$$

lets proceed with:

$$\begin{aligned} \int_{t_a}^{t_b} \dot{x}^2 &= \dot{x}x \Big|_{t_a}^{t_b} - \int_{t_a}^{t_b} \omega \dot{y}x \\ \dot{x}x \Big|_{t_a}^{t_b} &= \int_{t_a}^{t_b} (\dot{x}^2 + \omega \dot{y}x) dt \end{aligned} \quad (2.105)$$

Similarly,

$$\dot{y}y \Big|_{t_a}^{t_b} = \int_{t_a}^{t_b} (\dot{y}^2 - \omega \dot{x}y) dt \quad (2.106)$$

and,

$$\int_{t_a}^{t_b} \dot{z}^2 = \dot{z}^2 t \quad (2.107)$$

by substituting these values in equation of S_{cl} we get,

$$S_{cl} = \frac{m}{2} [\dot{x}x + \dot{y}y + \dot{z}^2 t]_{t_a}^{t_b} \quad (2.108)$$

Since the motion is on xy plane it must be cyclotronic. Therefore,

$$\begin{aligned} x &= x_0 + R \cos \omega t \\ y &= y_0 - R \sin \omega t \end{aligned} \quad (2.109)$$

$$\begin{aligned} \dot{x} &= -R\omega \sin \omega t \\ \dot{y} &= -R\omega \cos \omega t \end{aligned} \quad (2.110)$$

substituting these in S_{cl}

$$S_{cl} = \frac{m}{2} [-xR\omega \sin \omega t - yR\omega \cos \omega t + \dot{z}^2 t]_{t_a}^{t_b} \quad (2.111)$$

$$S_{cl} = \frac{m\omega}{2} \left[x(y - y_0) - y(x - x_0) + \frac{\dot{z}^2}{\omega} t \right]_{t_a}^{t_b} \quad (2.112)$$

or,

$$\begin{aligned} S_{cl} &= \frac{m\omega}{2} \left[-xy_0 + yx_0 + \frac{\dot{z}^2}{\omega} t \right]_{t_a}^{t_b} \\ S_{cl} &= \frac{m\omega}{2} \left[-(x_b - x_a)y_0 + (y_b - y_a)x_0 + \frac{(z_b - z_a)^2}{\omega(t_b - t_a)} \right] \end{aligned} \quad (2.113)$$

By satisfying the boundary condition i.e. at $t = t_a, x = x_a$ and at $t = t_b, x = x_b$, we get,

$$\begin{aligned} x_0 &= \frac{x_b + x_a}{2} + \frac{y_b - y_a}{\tan \frac{\omega T}{2}} \\ y_0 &= \frac{y_b + y_a}{2} - \frac{x_b - x_a}{\tan \frac{\omega T}{2}} \end{aligned} \quad (2.114)$$

Substituting (x_0, y_0) back to equation 2.113,

$$\begin{aligned} S_{cl} &= \frac{m}{2} \left[\frac{(z_b - z_a)^2}{T} + \frac{(\omega/2)}{\tan \frac{\omega T}{2}} ((x_b - x_a)^2 + (y_b - y_a)^2) \right. \\ &\quad \left. + \omega(x_a y_b - y_a x_b) \right] \end{aligned} \quad (2.115)$$

Now since Kernel is given by eq. 1.35,

$$K = F(T) e^{(i/\hbar) S_{cl}} \quad (2.116)$$

To find $F(T)$ For the motion in Z-axis it turns out to be same as that for free particle: i.e.

$$\sqrt{\frac{m}{2\pi\hbar i T}}$$

For the part corresponding to x,y plane we will proceed as,

$$\begin{aligned} K(0, 0, t + s; 0, 0, 0) &= \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy K(0, 0, 0, t + s; x, y, t) K(x, y, t; 0, 0, 0) \end{aligned} \quad (2.117)$$

or,

$$\begin{aligned} F(t + s) &= \\ F(t)F(s) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy \exp \left\{ \frac{im}{2\hbar} \left(\frac{(\omega/2)}{\tan \frac{\omega t}{2}} + \frac{(\omega/2)}{\tan \frac{\omega s}{2}} \right) (x^2 + y^2) \right\} \end{aligned} \quad (2.118)$$

Solving this we get:

$$\frac{F(t + s)}{F(t)F(s)} = \frac{4\pi\hbar i}{m\omega} \frac{1}{\frac{1}{\tan \frac{\omega s}{2}} + \frac{1}{\tan \frac{\omega t}{2}}} \quad (2.119)$$

On solving it we get,

$$\frac{F(t+s)}{F(t)F(s)} = \frac{2\pi\hbar i}{m\omega} \frac{2 \sin \omega t/2 \sin \omega s/2}{\sin \omega(t+s)/2} \quad (2.120)$$

$$F_{x,y}(t) = \left(\frac{m}{2\pi i \hbar} \right) \frac{\omega/2}{\sin \omega t/2} \quad (2.121)$$

Now together with Z part of $F(T)$ we get,

$$F(T) = \left(\frac{m}{2\pi i \hbar T} \right)^{3/2} \frac{(\omega T)/2}{\sin(\omega T)/2} \quad (2.122)$$

Chapter 3

Relating the Kernel to physical quantity

3.1 Kernel and Partition Function

In statistical mechanics Partition Function is given as:

$$\mathcal{Z} = \sum_n e^{-\beta E_n} \quad (3.1)$$

Where $\beta = \frac{1}{k_B T}$

It can be written as,

$$\mathcal{Z} = \langle n | \sum_n e^{-\beta H} | n \rangle \quad (3.2)$$

$$\Rightarrow \int dx \sum_n \langle n | x \rangle \langle x | e^{-\beta H} | n \rangle$$

$$\Rightarrow \int dx \sum_n \langle x | e^{-\beta H} | n \rangle \langle n | x \rangle$$

$$\Rightarrow \int dx \langle x | e^{-\beta H} | x \rangle = \text{Tr} e^{-\beta H} \quad (3.3)$$

$$\Rightarrow \int dx \langle x | e^{-\frac{i}{\hbar} H(-i\beta\hbar)} | x \rangle$$

$$\mathcal{Z} = \int dx K(x, x; -i\beta\hbar, 0) \quad (3.4)$$

Equation 3.4 gives us the relation between partition function $\mathcal{Z}$ and Kernel K .

3.2 Kernel and $1/H$

We can write H^{-1} as $\int_0^\infty dt e^{-Ht}$ Also we know that,

$$K(x, x, -i\hbar t, 0) = \langle x | e^{-Ht} | x \rangle$$

So we may write,

$$H^{-1} = \int_0^\infty dt K(x, x; -i\hbar t, 0) \quad (3.5)$$

3.3 Det(H) and Kernel

We will be using the following relations,

$$\ln(\det H) = \text{Tr} \ln(H) \quad (3.6)$$

$$\det(e^A) = e^{\text{Tr}(A)} \quad (3.7)$$

and,

$$-\text{Tr} \ln \left(\frac{a}{b} \right) = \int_0^\infty \frac{dt}{t} \text{Tr} (e^{-ta} - e^{-tb}) \quad (3.8)$$

Using eq. 3.3 we know,

$$\text{Tr} e^{-Ht} = \int dx \langle x | e^{-Ht} | x \rangle \quad (3.9)$$

$$\Rightarrow \int dx K(x, x; -i\hbar t, 0) \quad (3.10)$$

Using 3.8 and 3.6 we get,

$$-\text{Tr} \ln(H) = \int_0^\infty \frac{dt}{t} \text{Tr} (e^{-Ht} - e^{-t}) \quad (3.11)$$

$$-\det H = \int_0^\infty \frac{dt}{t} \text{Tr}(e^{-Ht}) - \int_0^\infty \frac{dt}{t} \text{Tr}(e^{-t}) \quad (3.12)$$

$$-\det H = \int_0^\infty \frac{dt}{t} \int dx K(x, x, -i\hbar t, 0) - \int_0^\infty \frac{dt}{t} \text{Tr}(e^{-t}) \quad (3.13)$$

3.4 Kernel and Ground State Energy

Since we know that kernel can be expressed as,

$$\begin{aligned} K(x_b, T; x_a, 0) &= \langle x_b, T | x_a, 0 \rangle \\ &= \langle x_b | e^{-\frac{i}{\hbar} H T} | x_a \rangle \end{aligned} \quad (3.14)$$

Now to find ground state energy we will use the euclidean path integral where we substitute time $T = -i\tau$ in above equation,

$$\begin{aligned} K(x_b, -i\tau; x_a, 0) &= \langle x_b, -i\tau | x_a, 0 \rangle \\ K_E(x_b, \tau; x_a, 0) &= \langle x_b | e^{-\frac{H}{\hbar} \tau} | x_a \rangle \end{aligned} \quad (3.15)$$

where K_E represents Euclidean path integral or euclidean kernel.

$$K_E(x_b, \tau; x_a, 0) = \langle x_b | e^{-\frac{H_n}{\hbar} \tau} | x_a \rangle \quad (3.16)$$

$$\begin{aligned} &= \sum_n \langle x_b | e^{-\frac{H_n}{\hbar} \tau} | n \rangle \langle n | x_a \rangle \\ &= \sum_n e^{-\frac{E_n}{\hbar} \tau} \langle x_b | n \rangle \langle n | x_a \rangle \end{aligned} \quad (3.17)$$

if $\tau \rightarrow \infty$ then maximum contribution to K_E will be from the minimum energy term (i.e. Ground State Energy E_0)

Therefore,

$$\lim_{\tau \rightarrow \infty} K_E = e^{-\frac{E_0}{\hbar} \tau} \langle x_b | 0 \rangle \langle 0 | x_a \rangle \quad (3.18)$$

$$\lim_{T \rightarrow \infty} \frac{\partial}{\partial T} K_E = -\frac{E_0}{\hbar} K_E \quad (3.19)$$

$$E_0 = -\hbar \lim_{\tau \rightarrow \infty} \frac{\partial}{\partial \tau} \ln K_E \quad (3.20)$$

Eq. 3.20 gives us the relation between ground state energy and kernel.

Chapter 4

Many-body Systems

4.1 System of harmonic oscillators on 1D Circular ring

For such a system lagrangian is given as:

$$\begin{aligned}\mathcal{L} &= \sum_j \frac{m\dot{x}_j^2}{2} - \frac{K}{2}(x_{j+1} - x_j)^2 \\ \Rightarrow \sum_j \frac{p_j^2}{2m} - \frac{K}{2}(x_{j+1} - x_j)^2\end{aligned}\quad (4.1)$$

By Fourier Transformation and Periodicity,

$$x_j = \sum_{l=0}^{N-1} \frac{\hat{x}_l}{\sqrt{N}} e^{\frac{2\pi i l j}{N}} \quad (4.2)$$

$$p_j = \sum_{l=0}^{N-1} \frac{\hat{p}_l}{\sqrt{N}} e^{\frac{2\pi i l j}{N}} \quad (4.3)$$

and Inverse fourier Transformation as,

$$\hat{x}_l = \sum_{j=0}^{N-1} \frac{x_j}{\sqrt{N}} e^{-\frac{2\pi i l j}{N}} \quad (4.4)$$

$$\hat{p}_l = \sum_{j=0}^{N-1} \frac{p_j}{\sqrt{N}} e^{-\frac{2\pi i l j}{N}} \quad (4.5)$$

Suppose mean position of j^{th} mass is ja , then total length of N oscillators is $Na = L$. Or we may write above equation as,

$$x_j = \sum_{l=0}^{N-1} \frac{\hat{x}_l}{\sqrt{N}} e^{\frac{2\pi i l j a}{Na}} = \sum_k \frac{\hat{x}_l}{\sqrt{N}} e^{ikja} \quad (4.6)$$

$$p_j = \sum_k \frac{\hat{p}_l}{\sqrt{N}} e^{ikja} \quad (4.7)$$

where $k = \frac{2\pi l}{Na} = \frac{2\pi l}{L}$, where $l \in \mathbb{Z}$ and goes from $l = 0$ to $l = N - 1$

Now using orthogonal property of transformation i.e.

$$\sum_j e^{-ik'ja} e^{ikja} = \sum_j e^{i(k-k')ja} = N\delta_{kk'} \quad (4.8)$$

Now the kinetic term of lagrangian,

$$\sum_j p_j^2 = \sum_j \left(\frac{1}{\sqrt{N}} \sum_k p_k e^{ikja} \right) \left(\frac{1}{\sqrt{N}} \sum_{k'} p_{k'} e^{ik'ja} \right) \quad (4.9)$$

$$\begin{aligned}\Rightarrow \sum_j \frac{1}{N} \sum_k \sum_{k'} p_k p_{k'} e^{i(k+k')ja} \\ \Rightarrow \sum_j \frac{1}{N} \sum_k \sum_{k'} p_k p_{k'} \delta_{k,-k'} = \sum_k p_k p_{-k}\end{aligned}\quad (4.10)$$

Since $p_{-k} = p_k^*$ and $p_k p_k^* = |p_k|^2$, therefore above equation can gives,

$$\sum_j p_j^2 = \sum_k |p_k|^2 \quad (4.11)$$

Now the potential term of lagrangian gives,

$$\Rightarrow \sum_j (x_{j+1} - x_j)^2 \quad (4.12)$$

By using equation 4.6, after fourier transformation it gives,

$$\begin{aligned}\sum_j (x_{j+1} - x_j)^2 \\ \Rightarrow \sum_j \frac{1}{N} \left(\sum_k (x_k e^{ik(j+1)a} - x_k e^{ik(j)a}) \right. \\ \left. \times \sum_{k'} (x_{k'} e^{ik'(j+1)a} - x_{k'} e^{ik'(j)a}) \right) \\ \Rightarrow \sum_j \frac{1}{N} \sum_k \sum_{k'} x_k x_{k'} e^{ik(j)a} e^{ik'(j)a} (e^{ika} - 1)(e^{ik'a} - 1) \\ \Rightarrow \sum_j \frac{1}{N} \sum_k \sum_{k'} x_k x_{k'} \delta_{k,-k'} (e^{ika} - 1)(e^{ik'a} - 1) \\ \Rightarrow \sum_k x_k x_{-k} (e^{ika} - 1)(e^{ik'a} - 1) \\ \sum_j (x_{j+1} - x_j)^2 = \sum_k |x_k|^2 \left(4 \sin^2 \frac{ka}{2} \right) \quad (4.13)\end{aligned}$$

therefore its lagrangian can be written as:

$$\mathcal{L} = \sum_k \frac{|p_k|^2}{2m} - \frac{K}{2} \left(4 \sin^2 \frac{ka}{2} \right) |x_k|^2 \quad (4.14)$$

If we assume some $\omega_k = 2\sqrt{\frac{K}{m}} \sin \frac{ka}{2}$ then we can rewrite lagrangian as,

$$\mathcal{L} = \sum_k \frac{|p_k|^2}{2m} - \frac{m}{2} \omega_k^2 |x_k|^2 \quad (4.15)$$

4.1.1 Quantization of this system

Hamiltonian for chain of oscillators is,

$$\mathcal{H} = \sum_k \frac{|p_k|^2}{2m} + \frac{m}{2} \omega_k^2 |x_k|^2 \quad (4.16)$$

Since this hamiltonian is decoupled it would have k pairs of creation and anihilation operator $(a_k^\dagger, a_k)$ which satisfies commutator relationship i.e.

$$[a_k^\dagger, a_{k'}^\dagger] = [a_k, a_{k'}] = \delta_{kk'} \quad (4.17)$$

$$(4.18)$$

and $(a_k^\dagger, a_k)$ are,

$$a_k = \sqrt{\frac{m\omega_k}{2\hbar}} \left(\hat{x}_k + \frac{i}{m\omega_k} \hat{p}_k \right) \quad (4.19)$$

$$a_k = \sqrt{\frac{m\omega_k}{2\hbar}} \left(\hat{x}_k - \frac{i}{m\omega_k} \hat{p}_k \right) \quad (4.20)$$

and since,

$$\begin{aligned} p_{k'}^\dagger &= p_{-k} \\ [x_k, p_{k'}^\dagger] &= i\hbar \delta_{kk'} \end{aligned} \quad (4.21)$$

Hamiltonian can be written in form of creation & anihilation operator as:

$$\mathcal{H} = \sum_k \left(a_k^\dagger a_k + \frac{1}{2} \right) \hbar \omega_k \quad (4.22)$$

and $a_k^\dagger a_k = \hat{N}$ so,

$$\mathcal{H} = \sum_k \left(N_k + \frac{1}{2} \right) \hbar \omega_k \quad (4.23)$$

which follows creation and anihilation i.e.,

$$[H_k, a_k] = -\hbar \omega_k a_k \quad (4.24)$$

$$[H_k, a_k^\dagger] = \hbar \omega_k a_k^\dagger \quad (4.25)$$

and they act on eigenstates as follows,

$$\begin{aligned} a|0\rangle &= 0 \\ a|n\rangle &= \sqrt{n}|n-1\rangle \end{aligned} \quad (4.26)$$

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (4.27)$$

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle \quad (4.28)$$

with ground state energy as,

$$E_0 = \frac{1}{2} \hbar \omega \quad (4.29)$$

Now for system of uncoupled oscillators we will have:

$$a_k^\dagger |n_1, n_2, \dots, n_k, \dots\rangle \propto |n_1, n_2, \dots, n_k + 1, \dots\rangle \quad (4.30)$$

$$a_k |n_1, n_2, \dots, n_k, \dots\rangle \propto |n_1, n_2, \dots, n_k - 1, \dots\rangle \quad (4.31)$$

$$\begin{aligned} |n_1, n_2, \dots, n_k, \dots\rangle &= \frac{1}{\sqrt{n_1! n_2! \dots n_n!}} \\ &\times \left((a_1^\dagger)^{n_1} (a_2^\dagger)^{n_2} \dots (a_N^\dagger)^{n_N} \right) |0, 0, \dots, 0\rangle \end{aligned} \quad (4.32)$$

where $|0, 0, \dots, 0\rangle = |0\rangle$ is vacuum state, or precisely,

$$|n_1, n_2, \dots, n_k, \dots\rangle = \prod_k \frac{1}{\sqrt{n_k!}} (a_k^\dagger)^{n_k} |0\rangle \quad (4.33)$$

4.1.2 Partition Function

We know that partition function for a given Hamiltonian $\mathcal{H}$ is given as:

$$\mathcal{Z} = \sum_{n_i \in \mathbb{Z}} e^{-\beta \mathcal{H}_{n_i}} \quad (4.34)$$

Therefore for Hamiltonian in equation eq. 4.23 Partition function is:

$$\mathcal{Z} = \prod_k \sum_{n_k=0}^{\infty} e^{-\beta(n_k + \frac{1}{2})\hbar\omega_k} \quad (4.35)$$

$$\mathcal{Z} = \prod_k Z_k \quad (4.36)$$

For single k^{th} decoupled oscillator sum over n_k gives:

$$Z_k = \sum_{n_k=0}^{\infty} e^{-\beta(n_k + \frac{1}{2})\hbar\omega_k} \quad (4.37)$$

$$Z_k = e^{-\beta \frac{\hbar\omega_k}{2}} \sum_{n_k=0}^{\infty} e^{-\beta n_k \hbar\omega_k} \quad (4.38)$$

where $\sum_{n_k=0}^{\infty} e^{-\beta n_k \hbar\omega_k} = \frac{1}{1 - e^{-\beta \hbar\omega_k}}$ which gives,

$$Z_k = \frac{e^{-\beta \frac{\hbar\omega_k}{2}}}{1 - e^{-\beta \hbar\omega_k}} \quad (4.39)$$

by substituting it back in equation 4.38 we get,

$$\mathcal{Z} = \prod_k \frac{e^{-\beta \frac{\hbar\omega_k}{2}}}{1 - e^{-\beta \hbar\omega_k}} \quad (4.40)$$

eq. 4.40 gives the partition function for this system.

Appendix A

Action Problems

Here we will be using equation 1.2 to find action integral for different cases in classical mechanics:

A.1 Free Particle Action

Free particle Lagrangian is given as $\mathcal{L} = \frac{m\dot{x}^2}{2}$

Using Euler Lagrange equation we get equation of motion as;

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad (\text{A.1})$$

$$\Rightarrow m\ddot{x} = 0 \quad (\text{A.2})$$

$$\Rightarrow \dot{x} = \left(\frac{x_b - x_a}{t_b - t_a} \right) \quad (\text{A.3})$$

Since action, $S = \int_{t_a}^{t_b} \mathcal{L} dt$

$$\Rightarrow S = \frac{m}{2} \frac{(x_b - x_a)^2}{(t_b - t_a)} \quad (\text{A.4})$$

A.2 Action for Harmonic Oscillator

Lagrangian for harmonic oscillator is given as:

$$\mathcal{L} = \frac{m}{2} (\dot{x}^2 - \omega^2 x^2) \quad (\text{A.5})$$

We may find action by putting $\mathcal{L}$ in equation 1.2

$$\Rightarrow S = \int \mathcal{L} dt = \int \left(\frac{m}{2} (\dot{x}^2 - \omega^2 x^2) \right) dt \quad (\text{A.6})$$

1st using equation 1.5 we can find equation of motion as:

$$\ddot{x} = -\omega^2 x \quad (\text{A.7})$$

Now lets find $\int \dot{x}^2 dt$ using integration by parts i.e.:

$$\Rightarrow \int \dot{x}^2 dt = x\dot{x}|_{t_a}^{t_b} - \int -\omega^2 x^2 dt$$

which can be re-written as:

$$\Rightarrow \int (\dot{x}^2 - \omega^2 x^2) dt = x\dot{x}|_{t_a}^{t_b} = S \quad (\text{A.8})$$

$$S = x\dot{x}|_{t_a}^{t_b} \quad (\text{A.9})$$

Now from equation of motion we may get x of form

$$x = A \sin(\omega t) + B \cos(\omega t) \quad (\text{A.10})$$

Let T be the time interval $t_a - t_b$, then at $t = 0$, $x = x_a$ and at $t = T$, $x = x_b$ Applying these boundary condition we will get A and B as:

$$B = x_a \quad (\text{A.11})$$

and

$$A = \left(\frac{x_b - x_a \cos(\omega T)}{\sin(\omega T)} \right) \quad (\text{A.12})$$

Using these values now we get x and $\dot{x}$ as :

$$x = \left(\frac{x_b - x_a \cos(\omega T)}{\sin(\omega T)} \right) \sin(\omega t) + x_a \cos(\omega t) \quad (\text{A.13})$$

and

$$\dot{x} = \left(\frac{x_b - x_a \cos(\omega T)}{\sin(\omega T)} \right) \cos(\omega t) - x_a \sin(\omega t) \quad (\text{A.14})$$

Now substituting these in equation A.9 we get action as:

$$S = x(\dot{x})|_0^T \quad (\text{A.15})$$

$$\Rightarrow S = x(T)\dot{x}(T) - x(0)\dot{x}(0) \quad (\text{A.16})$$

Solving this we would get classical action for Simple harmonic oscillator as:

$$S = \frac{m\omega}{2 \sin(\omega T)} [(x_b^2 + x_a^2) \cos(\omega T) - 2x_a x_b] \quad (\text{A.17})$$

A.3 Particle under a constant force F

For a particle moving under a constant force:

$$\mathcal{L} = \frac{m}{2} \dot{x}^2 + Fx \quad (\text{A.18})$$

From this using eq. 1.5 we get equation of motion as:

$$\ddot{x} = \frac{F}{m} \quad (\text{A.19})$$

Again for given boundary condition, let T be the time interval $t_a - t_b$, then at $t = 0$, $x = x_a$ and at $t = T$, $x = x_b$ From this we can get:

$$\dot{x} = \frac{F}{m}(t) + u_a \quad (\text{A.20})$$

where u_a is the initial velocity at $x = x_a$ and,

$$x = \frac{F}{2m}(t)^2 + u_a(t) + x_a \quad (\text{A.21})$$

Applying boundary conditions mentioned above we get initial velocity u_a as,

$$u_a = \frac{x_b - x_a}{T} - \frac{FT}{2m} \quad (\text{A.22})$$

Now by substituting x and $\dot{x}$ in equation 1.2 for given L we get,

$$S = \int \mathcal{L} dt = \int \left(\frac{m}{2} \dot{x}^2 + Fx \right) dt \quad (\text{A.23})$$

$$S = \int_{t_a}^{t_b} \left\{ \frac{m}{2} \left(\frac{F}{m}(t) + u_a \right)^2 + F \left(\frac{F}{2m}(t)^2 + u_a(t) + x_a \right) \right\} dt \quad (\text{A.24})$$

Solving this we get,

$$S = \frac{F^2}{m} \frac{T^3}{3} + u_a^2 T + F u_a T^2 \quad (\text{A.25})$$

Substituting u_a in this equation we get,

$$S = -\frac{F^2 T^3}{24m} + \frac{F}{2}(x_b - x_a)T + \frac{m}{2} \frac{(x_b - x_a)^2}{T} \quad (\text{A.26})$$

Summary

This project began with studying the path integral formulation and how the principle of least action in classical mechanics changes in quantum mechanics to include all possible paths. We then solved some simple quantum mechanics problems using both traditional and path integral methods, finding similar results.

Next, we looked at how the kernel relates to various physical observables. Finally, we examined a system of multiple oscillators, quantized it, and calculated its partition function. This approach demonstrated the usefulness and consistency of the path integral formulation in quantum mechanics.